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Project #58: Optimizing Stroke Care & Breaking Barriers: Achieving Rapid Door to Thrombolytic Times in a Freestanding ED

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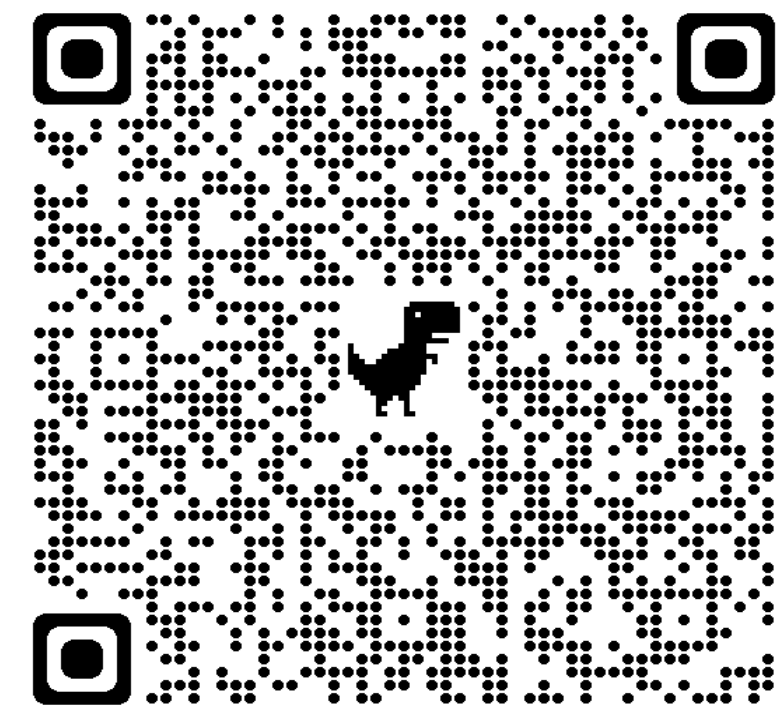
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Optimizing Stroke Care & Breaking Barriers: Achieving Rapid Door to Thrombolytic Times in a Freestanding ED



Team Members: Jacob Maine, Satheesh Gunaga, Darby Russell, Edyta Andersen, Ashley Krol, Toni Silas, Kristina Cross, Lin Andreoli, Timothy Manning, Laure Tartal, Holly Stos, Christina Sarkody, Ryan Spencer, Elizabeth Plemmons, Fadi Delly

Henry Ford Health Center Brownstown and Wyandotte Hospital – Department of Emergency Medicine, Wyandotte, MI

AIM

- 1. Primary Aim:** To establish a freestanding acute stroke ready certified Emergency Department at Henry Ford Health Center-Brownstown, serving as a proof-of-concept pilot for Henry Ford Health.
- 2. Early Stroke Identification and Treatment Aims:** Implement a redefined process to ensure that patients suffering from acute ischemic stroke in our community receive prompt, consistent, safe, and high-quality revascularization of at-risk brain tissue.
- 3. Implementation Aims:** Implement a robust monitoring and evaluation system to track compliance with the 15-15-15 stroke process and tools. Ensure comprehensive education for all ED staff regarding stroke processes and care goals.
- 4. Collaborative Aims:** Establish collaborative partnerships with neurology, radiology, ED and local EMS agencies to enhance stroke recognition and provide high quality streamlined care.

MULTI-DISCIPLINARY STROKE QI INITIATIVE IMPLEMENTATION & METHODOLOGY

- 1. Revised 15-15-15 Stroke Process:** Conducted re-education and clarification sessions to enhance understanding and reimplementation of the 15-15-15 stroke process, with a focus on achieving IV thrombolytic administration within 45 minutes of ED arrival.
- 2. Staff Education on Stroke Processes:** Implemented a comprehensive educational program for all staff members to ensure a clear understanding of stroke processes and time goals.
- 3. Enhanced Documentation Training:** Provided targeted education to ED providers, emphasizing the importance of clear and timely documentation, particularly related to delays in IV thrombolytic administration.
- 4. EMS Collaboration for Stroke Recognition:** Established a partnership with local EMS agencies to emphasize the significance of stroke recognition in the field and enhance the ED prenotification process.
- 5. PDCA Cycles by QI Team:** Implemented regular PDCA cycles led by the Quality Improvement (QI) team, involving tracking data, providing education, and delivering feedback to the care team on patients who received IV thrombolytics for acute stroke treatment.
- 6. Community Needs Assessment:** Conducted an assessment to identify the community's need for a stroke-designated freestanding emergency department, ensuring the provision of high-quality stroke care.
- 7. Community Outreach for Public Empowerment:** Engaged in community outreach initiatives to empower the public in recognizing stroke symptoms, encouraging them to call 911 promptly, and providing education on reducing stroke risk.

BROWNSTOWN ED ACHIEVED ACUTE STROKE READY HOSPITAL RECOGNITION

Henry Ford Brownstown ED

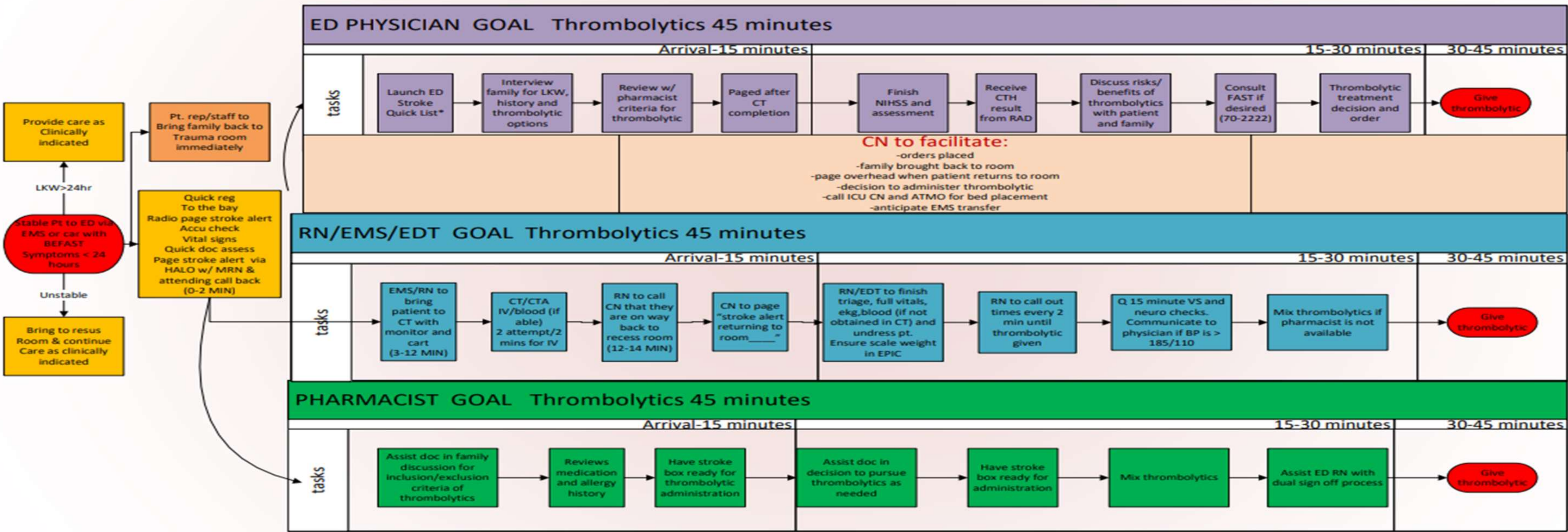
Brownstown ED Recieved Joint Commission Acute Stroke Ready Hospital Accreditation

STROKE

HF Brownstown Health Center is the First Free Standing ED to receive this Designation in the HFHS and in the State of Michigan.

CERTIFICATION
Meets standards for
Acute Stroke Ready Hospital

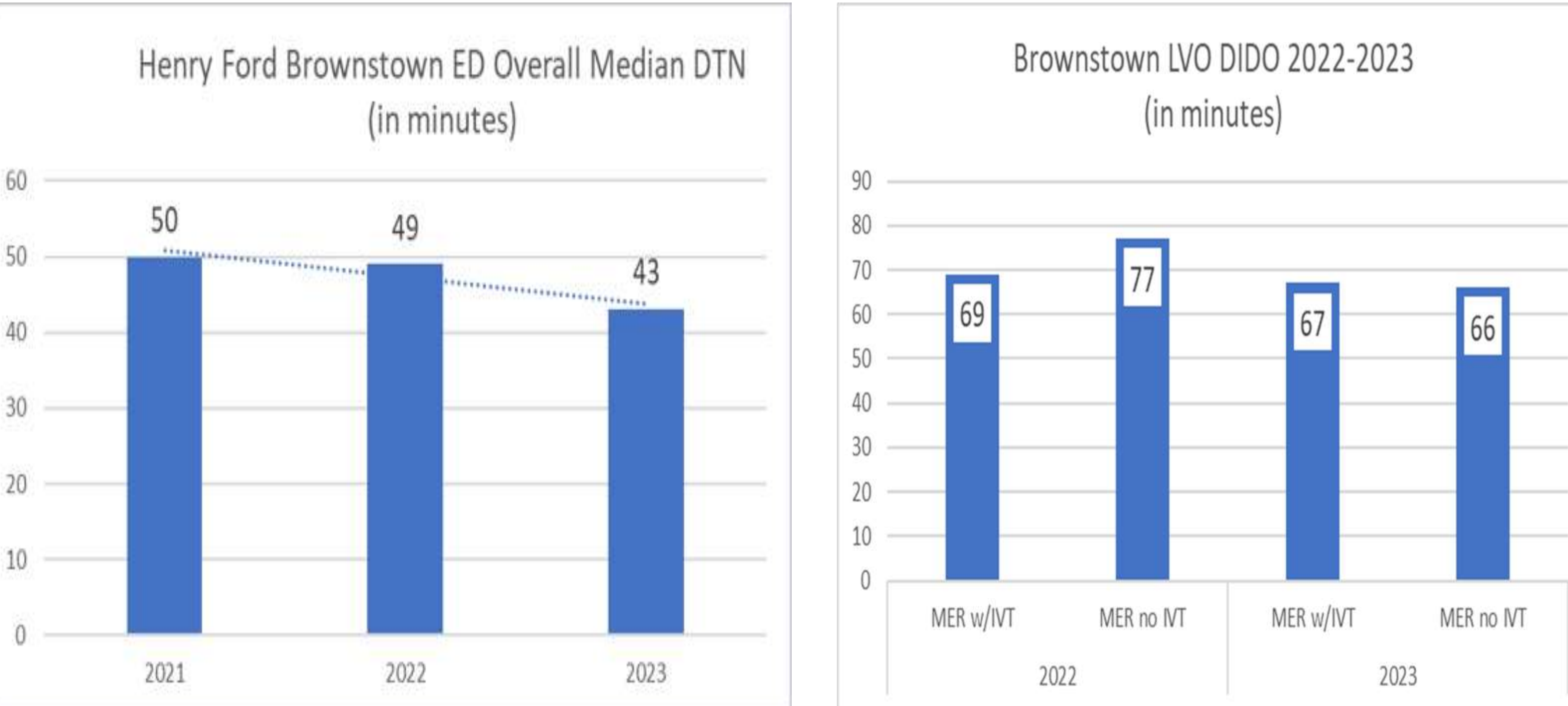
15-15-15 QI PROCESS & TOOLS : CHALLENGING THE 45 MINUTE THROMBOLYTIC BARRIER



DOOR TO NEEDLE TIMES : CHALLENGING THE 45 MINUTE THROMBOLYTIC BARRIER



ANNUAL DOOR TO NEEDLE & DOOR IN DOOR OUT TIMES FOR ACUTE ISCHEMIC STROKE



A3 LESSONS LEARNED

HF Brownstown Stroke Program Door to Needle (DTN) A3 2022-2023			
Project Name: Reducing Door to Needle (DTN) time in the treatment of acute ischemic stroke with the use of IV thrombolytics.	Sponsor: Stroke Committee	Team: HF Health Center Brownstown/Stroke Committee	Owners: Darby R., Edyta A., Ashley K., Toni S., Jacob M., Satheesh Gunaga.
Problem Background: The national benchmark that we are held to for DTN by the Joint Commission to maintain stroke certification is 60 minutes. To strive for the best stroke care possible, we have an internal goal of reaching a DTN of less than 45 minutes.	Implementation Plan: Starting in January 2022 we began to re-educate and re-emphasize our 15-15-15 process for IV thrombolytic stroke care. This dashboard is also utilized for PDCA cycles, process improvement and staff feedback. The 15-15-15 dashboard helps us to recognize where we are spending the most of our time.		
Current Condition: In 2022, we were able to reach our goal of 45-minute DTN with 30% compliance.	Results: In 2023, we have reached our goal. We have been able to achieve a DTN of less than 45 minutes at a compliance of 86%. [N of 15]		
Problem Analysis: Opportunities identified in first and second portions of our 15-15-15 stroke process. Delays noticed between patient arrival and return to department from CT. Delays also noticed between the return from CT and ordering of IV thrombolytic medication. These are gaps that require a multidisciplinary approach to reduce our times.	Effectiveness Check: Improvement seen in Door to CT overall from 14m to 9m and decreased variability of the CT completion to Decision process. Decision to needle reduction from 14 to 8 min. We will continue to review our 15-15-15 dashboard on a quarterly basis to monitor effectiveness. The dashboard will also be shared with Emergency Department staff on a quarterly basis to maintain awareness of our process.		
Target Condition: Target is to reach less than 45-minute DTN with 75% compliance. (Rationale: Every minute matters during an ischemic stroke. Odds of improved functional outcomes increases as time from onset of symptoms to IV thrombolytic therapy decreases)	Standardization: This process flow will be shared with other hospitals to help to positively impact the functional outcomes of patients suffering from acute ischemic stroke.		

PDCA LESSONS LEARNED

PLAN

- ✓ Coordinated Support from SLT
- ✓ Combined Resources from HFWH Stroke Team to support ASRH Efforts
- ✓ Developed 15-15-15 Tools
- ✓ Set DTN treatment time goals

ACT

- ✓ Analyze using 15-15-15 dashboard to determine future opportunities for improvement of DTN times
- ✓ Opportunity to standardize 15-15-15 stroke process across Henry Ford Health
- ✓ Standardized community stroke presentation to ensure consistent dissemination of stroke information

DO

- ✓ Role out and education of 15-15-15 stroke process with multidisciplinary team
- ✓ Continuity of education annually to ensure sustained knowledge and awareness
- ✓ Performing community stroke awareness presentations

CHECK

- ✓ Utilize 15-15-15 dashboard as an auditing tool to verify effectiveness of our process or any implementations
- ✓ Cipher rounding at Wyandotte Hospital to assess community educational needs
- ✓ Transfer times, dysphagia screening, hemorrhagic stroke care and various other measures are routinely tracked by the stroke team